

Machine-Brain Interface: A Giant Step for Mankind

As the brain-computer interface is being closely coupled with bionics, the medical field is witnessing a breakthrough, giving hopes to millions of amputated and paralysed people around the globe

ANAGHA P.

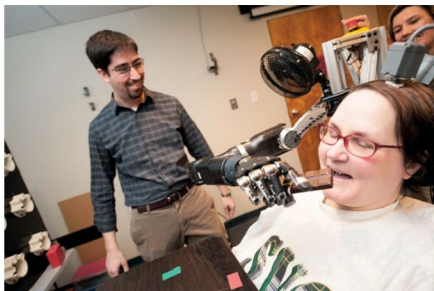
Jan Scheurmann was paralysed from neck down more than a decade ago due to degenerative brain disease. But thanks to the brain-computer interface (BCI), now she could grab and eat a chocolate with her robotic arm and even hi-five her doctors! With two aspirin-sized electrodes implanted to her brain and a few months of training, she was able to successfully manipulate a mind-controlled robot arm with seven axes of movement (front-back, up-down, left-right, wrist yaw, wrist pitch, wrist roll and hand grasp). After grabbing her first bite of chocolate she declared, "One small nibble for a woman, one giant bite for BCI."

"This is a spectacular leap towards greater function and independence for people who are unable to move their own arms," the senior investigator of the University of Pittsburgh's Pitt School of Medicine, Andrew Schwartz, said in a release. "This technology, which interprets brain signals to guide a robot arm, has enormous potential that we are continuing to explore. Our study has shown us that it is technically feasible to restore ability; the participants have told us that BCI gives them hope for the future."

In an earlier independent study by the BrainGate Implant, another paralysed woman, Cathy Hutchinson, was able to use the arm to pick up a thermos of coffee and drink it from a straw.

What is BCI

BCI, also known as a brain-machine interface (BMI), mind-machine inter-



Jan Scheurmann (Courtesy: <http://www.upmc.com>)

face (MMI) or direct neural interface, is a method by which the brain signals are used to control an external device. In simple terms, this technology helps you to control machines with thoughts. BCI has become one of the important areas of research in medical field as it can restore and augment human sensory, motor and cognitive functions.

The BCI system

A long-term goal of the BCI projects is to develop a neurotechnology that could convert thoughts into actions, thereby helping people with limb loss or paralysis to restore their movement and control, making them independent to a great extent.

The current BCI system comprises three main segments:

Sensor. A device implanted in the brain, usually multi-electrode arrays (MEAs), that records signals directly

related to imagined limb movement.

Decoder. A set of computers and specialised programs that can interpret the neural signals collected by the sensor, and which turn them to corresponding commands for an external device.

External device. A communication device like computer or a robotic limb on which the decoded brain signals are converted to required actions.

How it works

The first step is to associate raw neuronal signals with their corresponding movements. The magnetic resonance imaging mapping is used to find out the regions of the brain that are active when performing certain functions. This helps to place the electrode in the right target.

The sensors are smaller than a square centimetre and are implanted in the primary cortex of the brain. Each